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**Empirical Profiling of Cloud-Based Healthcare
Interoperability Gateways**

Master Thesis

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Abstract

The digitalisation of community-based healthcare requires heterogeneous applications to exchange information through regional services whose behaviour must be assessed across complete transaction paths. In the context of SI.Ter (Sistema Informativo Territoriale), the Territorial Information System for the Healthcare Organisations of the Veneto Region, this thesis defines an experimental methodology for end-to-end performance profiling and differential overhead characterisation of cloud-based healthcare interoperability gateways. Documented regional workflows become reproducible workloads observed at the client boundary. The Request Dataset Generator (RDG) is a Python framework that creates synthetic Fast Healthcare Interoperability Resources (FHIR) R4 (Release 4) patients, composes ordered multi-step workflows, schedules configurable traffic, and concurrently executes independent flow occurrences. It supports the direct RVE (Regione Veneto) patient-registry transactions RVE-54 and RVE-55 and a gateway-mediated workflow in mock and live modes. Structured logs provide a traceable dataset for request- and flow-level latency, throughput, error-rate, concurrency, and scheduling metrics. Since internal gateway logs remain outside the authorised perimeter, the *Surrogate Observability Strategy* uses the `direct_vs_middleware` experiment to compare client-observed latency distributions for direct and mediated paths. Differences in means and selected percentiles estimate apparent path-level overhead, not internal gateway time; routing, security, transformation, network, and retry effects may also contribute. The empirical campaign comprises 74 completed runs. As a proof of concept, it distinguishes transport from application outcomes, request from flow populations, direct from mediated paths, and successful from failed processing. The contribution is a profiling framework and a reproducible surrogate-observability protocol for constrained healthcare integration environments.

Contents

Acronyms and Abbreviations	viii
Glossary	ix
1 Problem Statement	1
1.1 Regulatory and Technical Context	1
1.1.1 Territorial Coordination	1
1.1.2 Heterogeneous Systems in SI.Ter	3
1.2 Continuity and Observability	8
1.2.1 Workflow Traceability	8
1.2.2 The Authorised Observation Boundary	10
1.3 Methodological Response	14
1.3.1 Surrogate Observability through RDG	14
1.3.2 Scope, Objectives, and Contributions	16
2 Solution Space	18
2.1 Scientific Basis of the Profiling Strategy	18
2.1.1 Experimental Alternatives and Validity	18
2.1.2 Synthetic Workflows as Experimental Instruments	20
2.2 Security-Constrained Observation Strategy	23
2.2.1 Authorised Observation Boundary	23
2.2.2 Client-Boundary Latency and Saturation Evidence	25
2.3 Healthcare Workflows as Information Dependencies	33
2.3.1 Protected Discharge and Anagrafe Zero	33
2.3.2 Cryptographic and Protocol Heterogeneity	37
2.4 Experimental Design and Evidence Semantics	40
2.4.1 Controls, Variables, and Reproducibility	40
2.4.2 Evidence Gates and Ex Ante Judgement	44
2.5 Generic versus Domain-Specific Profiling	46

CONTENTS

2.5.1	Modelling Distance and Semantic Fidelity	46
2.5.2	Ex Ante Justification of RDG	50
	Bibliography	i
	Online Sources	vi

List of Figures

1.1	Territorial-care operating model	2
1.2	RDG-orchestrated regional transactions	10
1.3	Surrogate observability pattern	15
2.1	Security-constrained surrogate observability	24
2.2	Network, transport, and application latency	26
2.3	Conceptual node load-state transitions	28
2.4	Conceptual Load-Sharing Window distribution	28
2.5	Point-to-point and collective communication patterns	30
2.6	Regional synchronous dependency chains	35
2.7	Persistent and stateless secure-connection FCT	43

List of Tables

1.1	Summary of key standards, including regulatory constraints, and their functional roles in the SI.Ter integration stack.	6
1.2	Comparison between internal production evidence and external experimental observation	12
2.1	Experimental alternatives and their admissible scientific role. . .	19
2.2	Operational measures and their role in saturation analysis. . . .	31
2.3	Information dependencies retained by the profiling model. . . .	36
2.4	Methodological separation of Scryba Sign profiling populations. .	39
2.5	Controls for an interpretable direct-versus-mediated differential. .	41
2.6	Ex ante criteria and required evidence.	44
2.7	Generic and domain-specific profiling paradigms	48

Source Code List

Acronyms and Abbreviations

APMS Application Profile Management System. [3](#)

ATNA Audit Trail and Node Authentication. [7](#), [11](#)

COT Centrale Operativa Territoriale. [1](#)

FHIR Fast Healthcare Interoperability Resources. [4](#)

FSE Fascicolo Sanitario Elettronico. [ix](#), [2](#), [7](#)

HL7 Health Level Seven. [16](#)

IAP Identity and Assertion Provider. [8](#)

IHE Integrating the Healthcare Enterprise. [5](#)

JSON JavaScript Object Notation. [4](#)

JSONL JSON Lines. [16](#)

JWT JSON Web Token. [6](#), [16](#)

RDG Request Dataset Generator. [10](#)

RVE Regione Veneto. [4](#), [15](#)

SAML Security Assertion Markup Language. [6](#), [16](#)

SI.Ter Sistema Informativo Territoriale. [3](#)

XDS Cross-Enterprise Document Sharing. [3](#)

Glossary

APMS A centralised system used by Azienda Zero to support user authentication and profile management for participating cloud-based applications.

[3](#)

ATNA An IHE integration profile that defines security measures for healthcare information systems, including node authentication, protected communications, and the generation and collection of audit records. [7](#), [11](#)

COT A territorial coordination centre introduced in the Italian healthcare model to coordinate transitions among hospital, community, home-care, and other territorial services. [1](#)

FHIR An HL7 standard for exchanging healthcare information electronically through modular resources, defined data models, and interoperable interfaces. [4](#)

FSE The Italian *Fascicolo Sanitario Elettronico (FSE)*_G infrastructure that makes health and social-care documents and data concerning an individual available to authorised actors. [2](#), [7](#)

HL7 A family of standards for the exchange, integration, sharing, and retrieval of electronic health information. [16](#)

IAP A regional security actor that authenticates a requesting application context and issues the identity assertion required by subsequent healthcare interoperability transactions. [8](#)

IHE An international initiative that publishes integration profiles describing how established healthcare standards can be used together to support specific interoperability use cases. [5](#)

JSON A text-based data-interchange format that represents structured data through objects, arrays, and primitive values. [4](#)

JSONL A line-oriented data format in which each line is an independent JSON value, allowing structured event and execution records to be processed incrementally. [16](#)

JWT A compact token format used to transmit claims between parties as a JSON object that can be digitally signed and, when required, encrypted. [6](#), [16](#)

RDG The software framework developed in this thesis to generate synthetic healthcare data, compose interoperability workflows, schedule workloads, execute requests, and collect structured experimental observations. [10](#)

RVE The acronym used as a prefix for interoperability transactions defined for the Veneto regional healthcare ecosystem. [4](#), [15](#)

SAML An XML-based standard for exchanging authentication and authorisation assertions between identity providers and service providers. [6](#), [16](#)

SI.Ter The Veneto regional initiative supporting territorial-care applications and their integration with local and regional healthcare systems. [3](#)

XDS and XDS.b An IHE integration profile for registering, discovering, and retrieving clinical documents across healthcare organisations. XDS.b denotes the revised profile variant used by the interoperability architecture considered in this thesis. [3](#)

Chapter 1

Problem Statement

1.1 Regulatory and Technical Context

1.1.1 Territorial Coordination

DM (Decreto Ministeriale) 77/2022 defines territorial care as a coordinated network of services rather than as a collection of isolated clinical encounters. Its organisational model includes the District, the Casa della Comunità, the COT (Centrale Operativa Territoriale), the 116117 operational centre, home-care services, community hospitals, and telemedicine-supported services¹. The decree therefore assigns digital information exchange an operational role: coordination across care settings depends on information that remains available when responsibility for a patient moves between hospital departments, territorial services, general practitioners, and home care. The COT makes this dependency explicit because it coordinates transitions among care settings and connects actors that belong to different organisational and technical domains. Its work requires current patient-identification data, care-transition information, clinical documentation, and communication with local and regional services². A protected discharge, for example, does not end when a hospital records the discharge decision: it continues through the transmission, assessment, assignment, and activation activities that make territorial care effective. The organisational process is consequently continuous even when the supporting applications are not.

¹Ministero della Salute. *Regolamento recante la definizione di modelli e standard per lo sviluppo dell'assistenza territoriale nel Servizio sanitario nazionale*. Ministerial decree Decreto 23 maggio 2022, n. 77. Italian Official Gazette, General Series no. 144, 22 June 2022. Ministero della Salute, May 23, 2022, Allegato 1, pp. 12, 20–21.

²[Ibid.](#), Allegato 1, pp. 20–21.

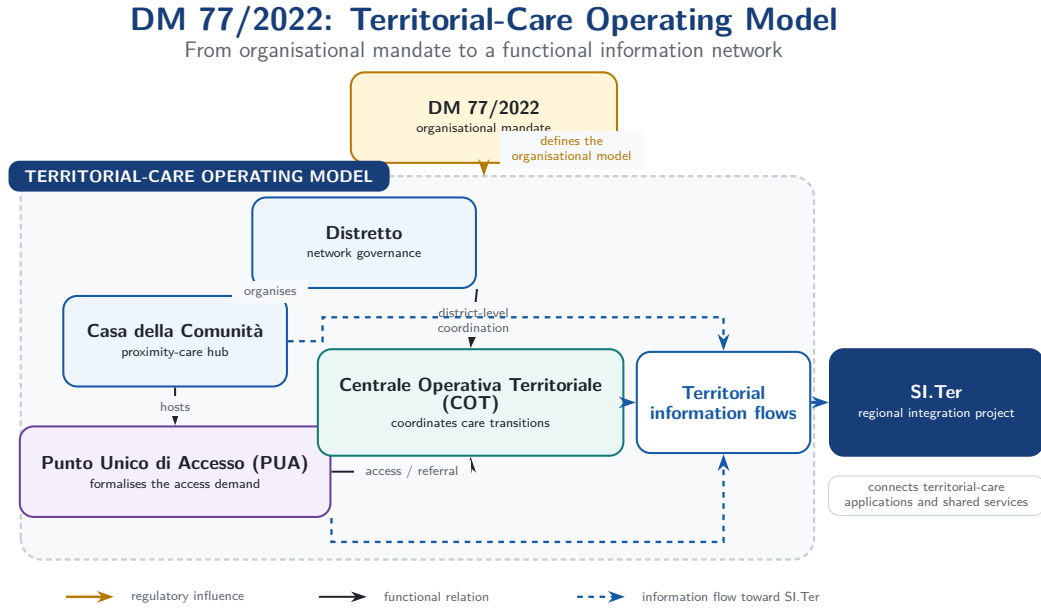


Figure 1.1: Concept of the territorial-care operating model defined by DM 77/2022.

This continuity requirement operates within a regulated data environment. Health data are a special category of personal data under Article 9 of European Union Regulation 2016/679, and their processing remains subject to data protection by design and by default and to measures appropriate to risk under Articles 25 and 32³. The FSE (Fascicolo Sanitario Elettronico), or Electronic Health Record, translates the same principle into role-based authorisation, periodic verification of access profiles, secure communication, operational traceability, and audit-log mechanisms⁴. DM 77/2022 thus creates a dual technical requirement: information must cross organisational boundaries to support the COT, yet each crossing must preserve identity, authorisation, confidentiality,

³European Parliament and Council of the European Union. *Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data*. Regulation (EU) 2016/679. Official Journal of the European Union, L 119. European Union, Apr. 27, 2016, pp. 1–88. URL: <https://eur-lex.europa.eu/eli/reg/2016/679/oj> (visited on 07/29/2026), arts. 9, 25 and 32.

⁴Ministero della Salute. *Fascicolo sanitario elettronico 2.0*. Ministerial decree Decreto 7 settembre 2023. Italian Official Gazette, General Series no. 249, 24 October 2023, as subsequently amended. Ministero della Salute, Sept. 7, 2023. URL: <https://www.gazzettaufficiale.it/eli/id/2023/10/24/23A05829/sg> (visited on 07/29/2026), art. 25.

and traceability. Satisfying both conditions requires the territorial-care model to interoperate with the heterogeneous systems already present in the regional ecosystem.

1.1.2 Heterogeneous Systems in SI.Ter

The regional programme provides the Veneto project context in which the organisational mandate becomes an integration problem. Regional documentation defines SI.Ter as a programme that connects territorial-care application domains, shared services, and pre-existing local and regional systems; it does not assume their replacement by a single greenfield platform⁵. The procurement framework accordingly requires cooperation across distinct ownership boundaries while preserving the processes and information assets already managed by Healthcare Organisations⁶. The COT model therefore depends on an architecture whose components remain heterogeneous in responsibility, interface, and data representation. The integration perimeter includes APMS (Application Profile Management System) for authentication and user-profile management in cloud-based applications⁷, Anagrafe Zero for patient identification and registration, local Admission, Discharge, and Transfer systems, and the FSEr (Fascicolo Sanitario Elettronico regionale) document infrastructure. Within the document path, XDS (Cross-Enterprise Document Sharing) repositories publish and retrieve clinical documents according to the regional Affinity Domain, while their indexing makes those documents available to the

⁵Regione del Veneto, Area Sanità e Sociale. *Modello di Governo per l'implementazione del Sistema Informativo Territoriale (SI.Ter) nelle Aziende ed Enti del Servizio Sanitario Regionale della Regione del Veneto*. Regional decree Decreto n. 11922. Regione del Veneto, Mar. 16, 2026, pp. 1–3.

⁶Azienda Zero. *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*. Technical specifications. Specific procurement, Digital Healthcare – Clinical-Care Information Systems, ID 2365, Lot 3; date derived from file metadata. Azienda Zero, July 9, 2025; Raggruppamento Temporaneo di Imprese guidato da AlmagivA. *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*. Technical report. Date derived from file metadata. Raggruppamento Temporaneo di Imprese guidato da AlmagivA, Sept. 19, 2025.

⁷DXC Technology. *Specifiche di integrazione autenticazione APMS*. Technical specification. Version 1.4.1. Azienda Zero, Nov. 22, 2024, Version 1.4.1, Section 1.

regional record. The documented SI.Ter architecture also connects context-call services, digital signature providers, prescription services, local clinical applications, and other shared components⁸. Within the integration stack represented by the thesis, Scryba Sign instantiates the signature-provider role and exposes its services through SOAP (Simple Object Access Protocol) interfaces described by WSDL (Web Services Description Language) contracts. The component handles FEA (Firma Elettronica Avanzata) and remote digital signatures; when the selected certificate and trust-service conditions satisfy the applicable eIDAS (electronic identification, authentication and trust services) requirements, the latter supports FEQ (Firma Elettronica Qualificata). It produces CAdES (CMS Advanced Electronic Signatures), PAdES (PDF Advanced Electronic Signatures), and XAdES (XML Advanced Electronic Signatures) envelopes and applies timestamping in the TimeStampedData (TSD) format. This classification identifies the supported integration surface and does not by itself certify the legal qualification of every configured signature⁹. The technological gap becomes visible in the interaction models exposed by the two principal data domains. Anagrafe Zero provides resource-oriented REST (Representational State Transfer) interactions over HTTP (Hypertext Transfer Protocol), based on HL7 (Health Level Seven) FHIR (Fast Healthcare Interoperability Resources) R4 (Release 4), for RVE (Regione Veneto) operations such as patient search and identifier assignment. FHIR admits both XML (Extensible Markup Language) and JSON (JavaScript Object Notation) representations, but the applicable regional interactions specify FHIR XML; the

⁸Raggruppamento Temporaneo di Imprese guidato da Almagora, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, Sections 2.1.2 and 2.1.4.

⁹Medas S.r.l. *Scryba Sign 3.x Developer's Guide*. Developer's guide. Version 16. Restricted internal document. Medas S.r.l., Apr. 10, 2024, pp. 7, 10, 28, and 89; European Parliament and Council of the European Union. *Regulation (EU) No 910/2014 on Electronic Identification and Trust Services for Electronic Transactions in the Internal Market*. Consolidated text of 18 October 2014. July 23, 2014. URL: <https://eur-lex.europa.eu/eli/reg/2014/910/2024-10-18/eng> (visited on 08/05/2026), Articles 3 and 25; Adriano Santoni. *RFC 5544: Syntax for Binding Documents with Time-Stamps*. RFC Editor. Feb. 2010. DOI: [10.17487/RFC5544](https://doi.org/10.17487/RFC5544). URL: <https://www.rfc-editor.org/info/rfc5544/> (visited on 08/05/2026), Section 2.

contrast is therefore not a simple XML-versus-JSON conversion¹⁰. The FSEr document infrastructure instead applies IHE (Integrating the Healthcare Enterprise) XDS.b transactions through SOAP (Simple Object Access Protocol) Web Services¹¹. Both families can use HTTP over TCP (Transmission Control Protocol) and protect the channel through mTLS (mutual Transport Layer Security). The mediation challenge therefore lies above this common transport substrate: SI.Ter must reconcile resource-oriented REST operations with message-oriented SOAP transactions and map patient identifiers, document metadata, security contexts, error semantics, and workflow dependencies without changing their clinical meaning. Interface connectivity alone is insufficient because syntactically valid forwarding does not guarantee semantic equivalence between the two application contracts. An end-to-end care transition can consequently traverse several systems before the receiving actor obtains the information required to continue the process. The solution architecture mediates these interactions through application interfaces and an interoperability layer. The layer mediates RESTful FHIR and SOAP-based IHE XDS.b interactions with ebXML (Electronic Business using eXtensible Markup Language) metadata, while managing asynchronous orchestration, transformation, errors, and interface lifecycles¹². Table 1.1 summarizes how the principal interoperability standards and regulatory constraints govern patient data, document exchange,

¹⁰Elena Vio et al. *Specifiche tecniche – Anagrafe Zero*. Technical specification. Version 2.6. Consorzio Arsenà.IT, Jan. 9, 2024, Sections 1.5.2 and 2.5.2.

¹¹IHE International. *IHE IT Infrastructure Technical Framework, Volume 1: Cross-Enterprise Document Sharing (XDS.b), Revision 20.2*. Nov. 11, 2025. URL: <https://profiles.ihe.net/ITI/TF/Volume1/ch-10.html> (visited on 07/22/2026); CNR (ICAR) e Dipartimento per la Trasformazione Digitale. *Specifiche tecniche per l'interoperabilità tra i sistemi regionali di FSE: Affinity Domain Italia*. Technical specification. Version 2.6.3. CNR (ICAR) e Dipartimento per la Trasformazione Digitale, Oct. 31, 2025.

¹²Raggruppamento Temporaneo di Imprese guidato da Al mavivA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, Sections 2.1.2 and 2.1.4.

access control, and auditability within the SI.Ter integration stack¹³¹⁴¹⁵.

Table 1.1: Summary of key standards, including regulatory constraints, and their functional roles in the SI.Ter integration stack.

Standard	Protocol or format	Role in the project
HL7 FHIR R4 Health Level Seven Fast Healthcare Interoperability Resources, Release 4	HTTP/REST; Patient and Bundle resources represented in FHIR XML, with JSON where required by the applicable interaction	Supports patient-registry interactions with Anagrafe Zero, including patient search, identifier assignment, and demographic updates such as RVE-54 and RVE-55.
IHE XDS.b Integrating the Healthcare Enterprise Cross-Enterprise Document Sharing	SOAP/XML messages, ebXML metadata, and the ITI (IT Infrastructure) transactions ITI-41, ITI-18, and ITI-43	Supports publication, metadata indexing and query, and retrieval of clinical documents through the FSEr Document Repository and Document Registry.
SAML (Security Assertion Markup Language) 2.0 / JWT (JSON Web Token)	XML assertions and compact RFC (Request for Comments) 7519 tokens carried by the transaction-specific SOAP or HTTP/REST exchanges	Establishes and propagates the security context: RVE-121 uses the SAML assertion to obtain a JWT, which is then used by RVE-130 for the contextual service call.

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¹³European Parliament and Council of the European Union, *Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data*, Article 25.

¹⁴Ministero della Salute, *Fascicolo sanitario elettronico 2.0*, Article 25.

¹⁵Azienda Zero, *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*, Section 4.5.

Standard	Protocol or format	Role in the project
GDPR (General Data Protection Regulation) / FSE (Fascicolo Sanitario Elettronico) 2.0	Data Protection by Design and by Default constraints	Establishes design constraints for differentiated access profiles and operation traceability. In the regional stack, IHE ATNA (Audit Trail and Node Authentication) implements audit logging. Authorisation checks and audit-record generation constitute non-bypassable processing in a compliant path; their cost therefore remains part of the measured end-to-end performance.
eIDAS; CAdES, PAdES, and XAdES Electronic-signature legal framework and signature-envelope profiles	SOAP Web Services; CMS, PDF, and XML signature envelopes; TimeStampedData temporal evidence	eIDAS supplies the legal classification of electronic signatures, while the envelope and timestamp formats define the Scryba Sign integration artefacts processed by the calling application.

These capabilities solve the connectivity problem only at the interface level. They do not by themselves establish whether a multi-system care process remains reconstructable from its initial request to its final outcome. Each mediation function introduces a boundary at which identifiers, security contexts, payloads, or timing information may change. Authentication can be a prerequisite for a patient query; the patient identifier returned by that query can become an input to document search; a retrieved document can affect a later care decision. The integration gap is therefore not the absence of interfaces, but the absence of a single observation point that represents the meaning, ordering, and outcome of all interactions involved in a care workflow. This gap turns the heterogeneity of SI.Ter into a problem of information continuity and end-to-end traceability.

1.2 Continuity and Observability

1.2.1 Workflow Traceability

Territorial care crosses organisational, application, and information boundaries at the same time. Organisational fragmentation arises because a single care pathway assigns responsibilities to hospital departments, the COT, territorial services, general practitioners, and home-care providers. Application fragmentation arises because these actors use software governed by different organisations. Information fragmentation arises because identity, clinical, administrative, and workflow data reside in registries, local applications, repositories, and regional platforms. Interoperability connects these domains, but the care process remains traceable only when the connections preserve the relationships among actors, data, and ordered activities. The protected-discharge workflow illustrates the resulting dependency chain. A hospital operator identifies the patient, assembles the information required for the referral, selects a proposed territorial setting, and submits the request to the competent COT. The COT receives and evaluates the request, coordinates the actors involved, and advances the case through explicit workflow states. The protected-admission workflow applies an analogous coordination pattern in the opposite direction, from a territorial setting towards a hospital structure¹⁶. These processes are multi-actor workflows rather than sequences of unrelated service calls: their correctness depends on the availability and interpretation of outputs produced by earlier activities.

The upper branch in Figure 1.2 makes the identity dependency concrete. Within `FLOW_PATIENT_QUERY`, RDG executes RVE-1.b before the RVE-54 Patient Query. Through the first transaction, the regional IAP (Identity and Assertion Provider) generates and returns a SAML 2.0 identity assertion; the execution engine extracts its required Base64 representation, stores it

¹⁶Raggruppamento Temporaneo di Imprese guidato da Al MAVIVA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, Sections 2.1.5 and 2.4.1.

in the context of the same flow occurrence, and resolves the corresponding placeholder in the **Authorization** header of RVE-54 before sending the query. The resulting header carries the assertion with the **IHE-SAML** authentication scheme, as required by the regional security profile¹⁷¹⁸. RDG therefore represents regional security-session state as explicit per-occurrence dependency data rather than reducing it to the state of the underlying HTTP connection. If RVE-1.b fails, the required assertion cannot be extracted, or its placeholder remains unresolved, the engine does not execute a semantically valid patient query and records the workflow as incomplete. This behaviour makes assertion acquisition and propagation part of the tested workflow. An isolated request can exercise RVE-54 only after receiving a valid security context from outside the test and therefore cannot evaluate the identity hand-off on which the regional interaction depends.

A technical evaluation based on isolated endpoints cannot represent this dependency. A successful authentication request does not prove that the resulting security context is accepted by the next service; a successful patient query does not prove that the returned identifier reaches document search; and a set of individually successful calls does not prove that the workflow reaches its intended terminal state. The relevant unit of analysis is therefore one occurrence of an end-to-end workflow, considered both through its executed interactions and through its overall outcome and completion time. This unit preserves the functional meaning that disappears when measurements are aggregated only by endpoint. End-to-end reconstruction also requires stable correlation identifiers, ordered timestamps, transport and application outcomes, and explicit data dependencies between steps. Without this evidence, a failure can be located at the client boundary but cannot be associated reliably with the workflow state that precedes or follows it. The continuity problem consequently becomes an

¹⁷Mauro Zanardini et al. *Infrastruttura di sicurezza GDL-O Sicurezza*. Technical specification. Version 2.14. Consorzio Arsenal.IT, June 28, 2024, Section 4.2.

¹⁸Vio et al., *Specifiche tecniche – Anagrafe Zero*, Section 1.7.

observability problem: the evaluation method must reconstruct the external behaviour of the distributed workflow, while the regional security perimeter limits which internal evidence the experiment can inspect.

Figure 1.2 maps this requirement to the Request Dataset Generator (RDG) flow model, in which each correlated occurrence retains both business outputs and the security context required by downstream transactions.

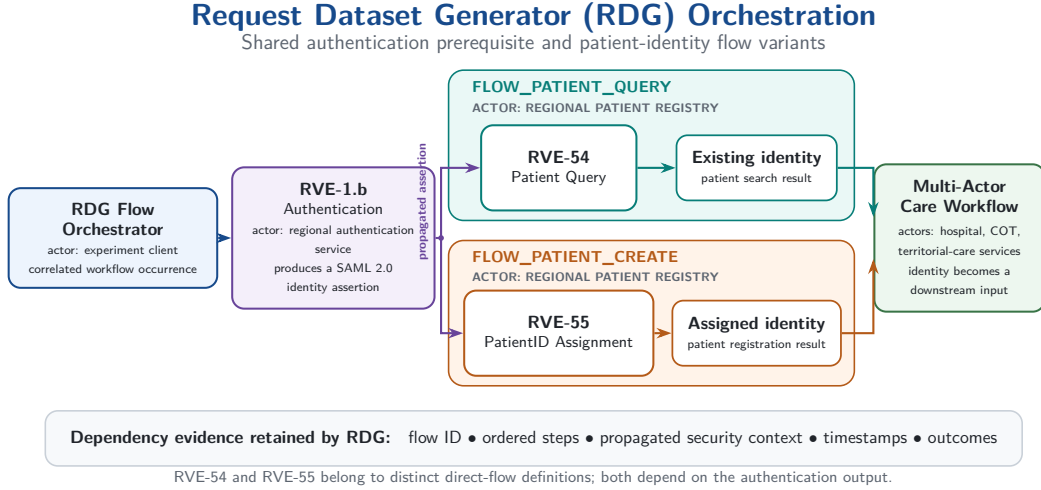


Figure 1.2: RDG orchestration of the RVE-1.b authentication output into regional patient-identity flows.

1.2.2 The Authorised Observation Boundary

Regional security specifications make authentication, authorisation, secure transport, and auditability integral to the integration architecture. Operators and applications act within an authenticated and authorised context, and assertions or tokens can become prerequisites for later services¹⁹. The architecture also provides audit mechanisms for production accountability; the regional security specification states that participating actors generate audit records and refers their detailed structure to the applicable ATNA (Audit

¹⁹DXC Technology, *Specifiche di integrazione autenticazione APMS*, Version 1.4.1; Zannardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*; Gregorio Canal and Gianluca Pavan. *Specifiche tecniche chiamata di contesto RVE*. Technical specification. Version 1.3. Consorzio Arsenà.IT, July 18, 2025.

Trail and Node Authentication) profile²⁰. The existence of this production telemetry does not make it part of the evidence available to every consumer or experiment, because access remains governed by operational roles and security policy. Within the authorised perimeter of this thesis, internal APMS telemetry and telemetry from the mediated gateway path are not made available to the experiment. This statement describes the research boundary; it does not claim that the platforms fail to produce logs or that regional policy universally prohibits their inspection. Real patient data, credentials, private endpoints, confidential deployment parameters, and production audit records likewise remain outside the disclosed dataset. Production services are neither instrumented by the thesis nor subjected to intrusive load.

These restrictions, summarised in Table 1.2, preserve the applicable security boundary but remove the internal timestamps required for a causal decomposition of gateway processing. The observable interval consequently extends from the client sending a request to the client receiving the corresponding response. It can include client-side processing, authentication, network transit, gateway mediation, downstream service execution, and response propagation. A latency measured across this interval characterises the path as a whole; it cannot attribute a delay to a specific gateway component without additional internal evidence. Treating the entire difference as gateway processing therefore exceeds the available data and violates the distinction between observation and causal explanation. This limitation determines the methodological requirement. The experiment needs a controlled external layer that represents complete workflows, fixes the workload and observation rules, records client-visible events, and compares direct with gateway-mediated execution at the same declared boundary. Such a layer does not replace production telemetry. It provides surrogate evidence for the effect visible outside the protected platform and makes every inference conditional on the comparability of the observed paths. The resulting research question is: *how can documented healthcare interop-*

²⁰Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*, Section 4.3.15.

erability workflows be translated into reproducible, observable, and measurable end-to-end workloads, and how can the apparent overhead associated with gateway mediation be estimated differentially when internal platform telemetry is outside the authorised observation perimeter?

Table 1.2: Comparison between internal production evidence and external experimental observation.

Dimension	Authorised Observation Boundary (Internal)	Experimental Observation Perimeter (RDG)
Data access	Real clinical data and ATNA audit records remain production-side and are accessible only to authorised operational roles.	RDG uses synthetic FHIR patients; real health data and production audit records remain outside the dataset.
Granularity	Component-level timestamps and traces remain protected; their availability depends on production instrumentation and authorised operator access.	RDG measures client-observed mediated-path latency $L_{\text{med}}(q)$ and the differential $\Delta L(q) = L_{\text{med}}(q) - L_{\text{dir}}(q)$ for the declared statistic q .
Outcome insight	Internal traces can associate a failure with a component only when the corresponding production evidence is available.	RDG distinguishes the HTTP transport outcome from application completion and semantic rejection visible in the response or output extraction.

Continued on the next page

Dimension	Authorised Observation Boundary (Internal)	Experimental Observation Perimeter (RDG)
Telemetry	APMS, mediated-gateway, and production ATNA logs remain inaccessible to the experiment.	RDG produces correlated JSONL events for requests, responses, and completed workflow occurrences.
Overhead attribution	Direct gateway measurement requires aligned ingress, internal-component, and egress timestamps, which lie outside the experimental perimeter.	ΔL estimates an aggregate path-level effect that can include routing, security, transformation, downstream-service, and retry costs; it does not decompose APMS or gateway processing.

1.3 Methodological Response

1.3.1 Surrogate Observability through RDG

RDG answers this question as the technical instrument of a *Surrogate Observability Strategy*. The strategy translates documented healthcare interactions into synthetic and executable workloads, observes them at the client boundary, and compares controlled direct and gateway-mediated paths. Its central differential mechanism is the `direct_vs_middleware` experiment. The current implementation groups direct patient-query and patient-creation flows, contrasts their client-observed end-to-end completion-time distributions with the mediated patient-query flow, and reports differences for the mean and the 50th, 95th, and 99th percentiles. These differences estimate *apparent path-level overhead*; they do not decompose the internal execution time of the gateway. Observation in this strategy is not synonymous with measuring network latency. The client-observed interval covers the complete request–response path, while RDG records the HTTP-level outcome and the application outcome as distinct evidence. An HTTP status code 200 confirms that the exchange returns a successful HTTP response, but it does not by itself prove that the payload contains the required output or that the workflow can advance. The execution engine therefore marks a step as failed when a mandatory output cannot be extracted, even if the response status is 200, and propagates that failure to the flow-level outcome. The same observation model retains service-reported semantic rejections, such as an RVE-55 status code 422 associated with FHIR validation, without inferring the specific violated constraint when payload-level evidence is unavailable. The sanitised evidence set illustrates the first distinction through RVE-TOKEN responses that carry status code 200 but remain application failures because the required token is not extracted. RVE-121 provides a complementary limit: its observed status code 200 demonstrates the current HTTP exchange, whereas evidence item E-RVE121-001 does not establish that a specific correction causes a transition from an earlier status code

400 to 200. Surrogate observability therefore separates HTTP-level evidence, application completion, and causal claims instead of treating every successful HTTP response as an equivalent logical outcome.

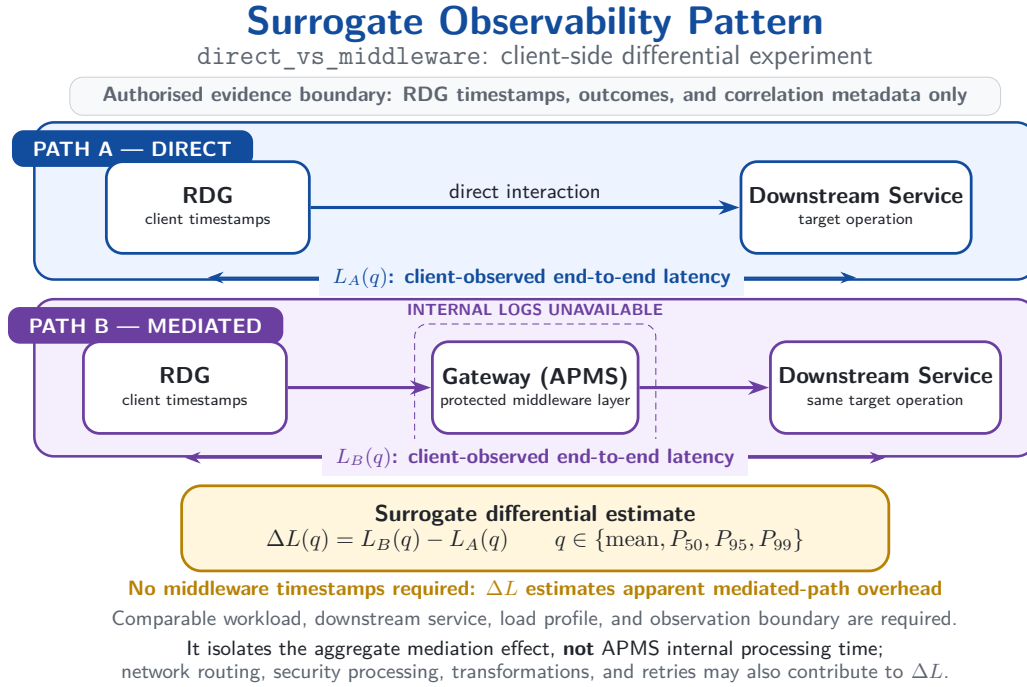


Figure 1.3: Surrogate observability pattern for `direct_vs_middleware` experiment.

The framework preserves the process semantics required by that comparison. It generates synthetic FHIR R4 patients, composes selected RVE (Regione Veneto) transactions into ordered multi-step flows, resolves outputs that become inputs to later steps, and assigns traffic occurrences to configurable timelines. Within each workflow occurrence, dependent steps execute sequentially; independent occurrences can execute concurrently through Python `asyncio`. Structured client-side events retain flow identity, step identity, timing, transport status, application outcome, and scheduling context. The resulting observation layer reconnects the functional workflow described by the SI.Ter documentation with the measurements available outside the protected gateway. The strategy covers selected workflow families related to regional authentication, patient identity, application context, document exchange, and digital

signature. Its standards perimeter includes HL7 (Health Level Seven) FHIR R4 resources, RVE specifications, IHE XDS.b transactions, SAML (Security Assertion Markup Language) assertions, and JWT (JSON Web Token) tokens. Synthetic patients replace real health information, while mock and authorised live modes separate controlled verification of the pipeline from observation of remote paths. This scope supports a reproducible performance method without claiming clinical effectiveness, legal certification, exhaustive coverage of SI.Ter, or successful validation of every implemented transaction.

1.3.2 Scope, Objectives, and Contributions

The method pursues four verifiable objectives:

1. interpret the standards, protocols, security mechanisms, and integration architectures that define the selected healthcare workflows;
2. formalise protected discharge, protected admission, and the associated interoperability interactions in terms of actors, preconditions, ordered steps, and information dependencies;
3. provide a configurable framework that translates those dependencies into synthetic, executable, and observable workloads;
4. define measurements that support differential comparison between direct and gateway-mediated paths.

The objectives establish an evidence-producing method and artefact; they neither promise a performance improvement nor predetermine the experimental outcome. RDG makes the objectives inspectable through a run-specific artefact chain. The chain comprises a resolved configuration, a synthetic dataset, a traffic timeline, a JSONL (JSON Lines) execution log, a normalised analytical dataset, metric reports, tabular summaries, and optional graphs. The analytical layer separates request latency, workflow completion time, throughput, error rate, concurrency, scheduling delay, and log completeness. Artefact presence

establishes software output coverage, whereas only an executed and quality-checked scenario provides experimental evidence. This distinction keeps the methodological contribution independent from any unsupported claim about observed performance. The contribution is consequently a reproducible procedure for profiling healthcare interoperability paths when the internal gateway remains outside the observation perimeter. The analytical component reconstructs the meaning and dependencies of the selected regional flows; the software component generates, executes, correlates, and measures their occurrences; the differential component quantifies the externally visible difference between declared direct and mediated paths. Together, these components close the causal chain that begins with the coordination mandate of DM 77/2022: the COT requires cross-setting information continuity, SI.Ter realises that requirement through heterogeneous integrations, distributed workflows create an end-to-end traceability problem, security constrains internal observation, and RDG supplies the surrogate evidence needed for controlled evaluation. The organisation of the thesis preserves the same progression. Chapter 2 examines the architectural, interoperability, workload, and performance foundations required to interpret the solution space. It also states the requirements, use cases, transactions, and ex ante evaluation criteria supported by the available sources. Chapter 3 presents RDG from the inside: framework design and implementation, data and workload generation, execution and observability, the experimental protocol, and the results produced by declared runs. Chapter 4 closes the traceability chain from objectives and artefacts to criteria, key performance indicators, and results; it assesses the contribution, states the limits of the evidence, and derives future directions from the actual findings.

Solution Space

2.1 Scientific Basis of the Profiling Strategy

2.1.1 Experimental Alternatives and Validity

The methodological problem is not the generation of a large number of requests. It is the production of interpretable performance evidence for stateful healthcare workflows when the experimenter controls the initiating client but does not control the complete regional execution path. The relevant unit of analysis is therefore an information-dependent workflow rather than an isolated endpoint. A valid experiment preserves the order in which security material, patient identifiers, context information and application outcomes become available. It also distinguishes the load offered to the system from the work that the system completes.

Four experimental strategies define the solution space. Operational observation offers high ecological fidelity, but it does not isolate the offered load from concurrent changes in users, services, routing, and infrastructure. An isolated protocol benchmark offers strong repeatability, but it removes the state transitions that determine whether a regional workflow reaches its terminal outcome. Trace replay can preserve an observed temporal pattern, but it requires an authorised trace whose state, ordering, and dependencies support faithful reconstruction. Controlled synthetic workflows make composition, timing, concurrency, and randomisation explicit, but their resemblance to production traffic requires independent validation.

Zhu, Chen, and Chiueh provide the decisive methodological distinction. Their trace-replay study treats the input trace, the initial system state, and the dependencies among operations as necessary conditions for accurate replay. The authors also show that temporal or spatial scaling remains credible only when

it does not violate those dependencies¹. Their result does not establish the universal superiority of replay over synthetic benchmarks. It establishes that replay validity depends on evidence that exists before the experiment begins. In the regional case examined by this thesis, no authorised production trace, complete state reconstruction, sanitisation procedure, or replay agreement enters the available evidence perimeter. Consequently, trace replay is not an admissible experimental option. Within this perimeter, controlled synthetic workflows constitute the only strategy that simultaneously preserves internal validity, data minimisation, and repeatability. The conclusion is conditional rather than absolute: synthetic execution is the only controllable route among the evidence sources that the experimenter is authorised to use. It does not reproduce an unknown operational distribution by assumption. Instead, it creates matched experimental populations whose differences follow from declared factors. This distinction prevents the absence of production evidence from becoming an implicit claim of production representativeness.

Table 2.1 separates the scientific role of each alternative from the claim that it can support.

Table 2.1: Experimental alternatives and their admissible scientific role.

Strategy	Primary strength	Admissible claim
Operational observation	Natural service mix and environmental realism.	Descriptive evidence for the observed environment when access and provenance are complete.

Continued on the next page

¹Ningning Zhu, Jiawu Chen, and Tzi-cker Chiueh. “TBBT: Scalable and Accurate Trace Replay for File Server Evaluation”. In: *4th USENIX Conference on File and Storage Technologies (FAST 05)*. San Francisco, CA: USENIX Association, Dec. 2005. URL: <https://www.usenix.org/conference/fast-05/tbbt-scalable-and-accurate-trace-replay-file-server-evaluation> (visited on 07/29/2026).

Strategy	Primary strength	Admissible claim
Isolated protocol benchmark	Repeatable control of one exchange.	Baseline for one contract, not completion of a stateful healthcare flow.
Trace replay	Preservation of captured ordering and timing when state and dependencies remain reconstructable.	Reproduction of an authorised trace within its capture and scaling assumptions.
Synthetic workflow	Explicit flow mix, timing, seed, concurrency, and data policy.	Internally valid comparison of declared scenarios; external representativeness remains unproven.

The resulting validity model separates four questions. Construct validity concerns whether the executable workload preserves the documented meaning and prerequisites of each transaction. Internal validity concerns whether matched scenarios differ only in the intended factor. External validity concerns whether an independent regional source supports the configured temporal profile, payload classes, and flow mix. Conclusion validity concerns whether repeated observations, explicit populations, and visible missing data support the reported comparison. Reproducibility therefore does not imply representativeness, and a realistic observation does not imply causality when uncontrolled environmental changes coexist with the measured path.

2.1.2 Synthetic Workflows as Experimental Instruments

A synthetic workflow has scientific value only when it acts as an experimental instrument rather than as a collection of fabricated requests. Its structure derives from authoritative transaction and process specifications; its data remain

synthetic; its schedule derives from declared stochastic or deterministic rules; and its outcomes remain classified at transport, application, and complete-flow levels. These properties make the independent variables inspectable and the dependent variables reproducible without requiring real patient records or security artefacts.

The strategy distinguishes *what* executes from *when* execution starts. The flow mix specifies semantically distinct workflows. The timeline assigns their planned start times. A homogeneous Poisson process supplies one reproducible baseline. For intensity $\lambda > 0$, the arrival count in an interval of duration t follows

$$\Pr[N(t) = k] = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k = 0, 1, 2, \dots \quad (2.1)$$

A non-stationary profile replaces the constant rate with $\lambda(t)$, for which

$$\mathbb{E}[N(a, b)] = \int_a^b \lambda(u) du. \quad (2.2)$$

Piecewise rates express controlled ramps and peaks, while parameterised bursts express short concentrations around a baseline. These models define experimental hypotheses; they do not estimate Veneto clinical demand in the absence of an authorised operational series.

The synthetic strategy preserves a second separation between flow arrivals and request arrivals. One flow can expand into different numbers of requests, carry different payload classes, and terminate at different points. Equal flow rates therefore do not imply equal request rates or equal remote work. A shift toward document processing changes transferred bytes and downstream computation; a shift toward authentication and context access changes the number of dependent steps and the probability of early termination. The experiment reports both the configured flow composition and the realised transaction composition, because a load comparison without this information confounds intensity with workload content.

Synthetic data also perform a methodological role. Metadata-driven generation

creates structurally and semantically valid patient and document contexts without using personal data. Stateful orchestration associates every derived identifier, assertion, token, or document reference with one flow occurrence, while a session-aware identity hand-off prevents security material from crossing concurrent occurrences. The asynchronous execution model permits concurrency among independent occurrences while preserving sequential dependencies inside each occurrence. These architectural properties support controlled concurrency without weakening the semantics of the workflow.

The design includes an explicit calibration condition. A controlled local path establishes the range in which scheduling, event production, and analysis sustain the intended offered load without material drift. This calibration does not predict the capacity of a regional service. It determines whether the workload generator remains a credible experimental instrument. When delivery drift already increases in the control condition, a corresponding live scenario cannot support a conclusion about external saturation.

The statistical unit follows the same hierarchy. Requests within one flow share state, flows within one run share workload and environmental conditions, and repetitions provide the highest-level comparable units. Request distributions remain useful for transaction analysis, but the method does not treat every request as an independent replicate. This rule prevents a large sample count from overstating the amount of independent experimental evidence.

The selected strategy consequently favours internal validity over unsupported realism. It makes every assumption visible: flow weights, arrival profile, payload class, connection policy, seed, concurrency bound, execution mode, and outcome rule. An independent authorised traffic source can later test external validity without changing the measurement semantics. Until such a source exists, the thesis presents synthetic workloads as controlled experimental inputs and never as reconstructed regional demand.

2.2 Security-Constrained Observation Strategy

2.2.1 Authorised Observation Boundary

The Authorised Observation Boundary originates from regional security, privacy, and organisational control, not from a limitation of the workload generator. The Request Dataset Generator (RDG) can initiate approved synthetic traffic and retain sanitised client-side events. The experimenter cannot inspect private execution phases inside the Application Profile Management System (APMS), regional middleware, or downstream services, and cannot retain assertions, access tokens, credentials, certificate secrets, private endpoints, or personal data. The boundary therefore determines which causal statements the evidence permits².

This constraint leads to an *Advanced Black-box with Surrogate Observability* methodology. The black-box component treats the protected regional path as an externally observable service. The advanced component preserves more than a request timestamp: it observes planned and actual dispatch, transport and application outcomes, dependency propagation, terminal flow state, flow completion time, scheduling drift, concurrency, and matched path differentials. The surrogate component uses these correlated external signals to characterise the path without pretending to expose its internal phases.

The boundary changes the meaning of overhead. A mediated request can include authentication, authorisation, envelope handling, semantic validation, transformation, routing, queueing, upstream waiting, and backend processing. A client-side duration contains their composite effect together with network and client contributions. It does not equal gateway processing time, and a slower mediated response does not identify one internal component as its cause. Optional authorised internal timestamps can validate or refine the external estimate, but their absence limits causal attribution rather than invalidating the

²DXC Technology, *Specifiche di integrazione autenticazione APMS*; Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*.

matched external comparison.

Figure 2.1 expresses this methodological boundary. The internal functions remain analytically plausible contributors, but the experiment measures only their aggregate path effect.

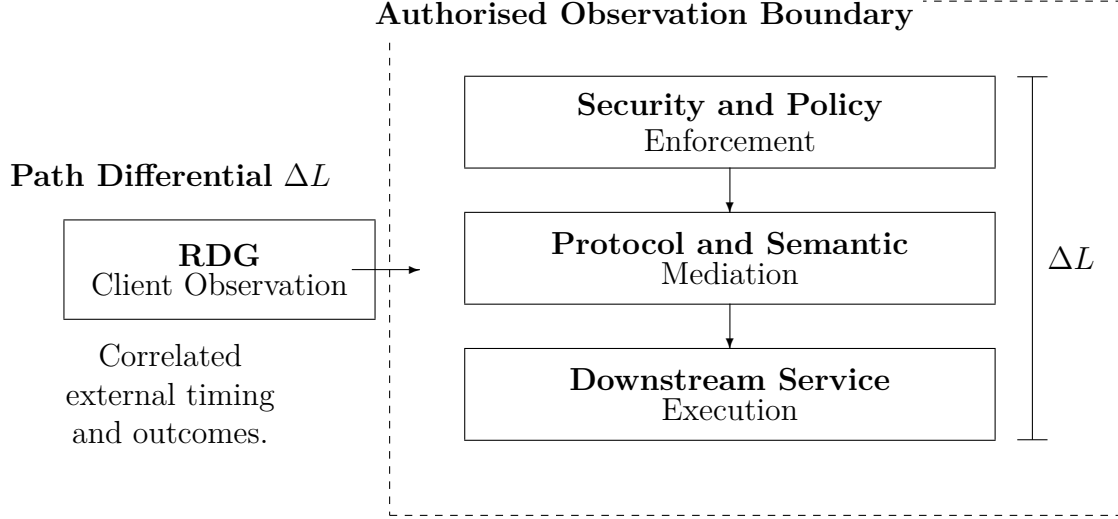


Figure 2.1: Security-constrained surrogate observability. RDG observes the aggregate external path while internal security, mediation, and service phases remain outside the authorised evidence perimeter.

Security state remains part of the workload even when its value remains absent from the evidence. Reusing one assertion or token across unrelated occurrences can increase apparent throughput while violating the transaction contract and removing a mandatory acquisition cost. The method therefore scopes security outputs to one occurrence and records only whether their production and propagation succeed. Data minimisation removes protected values, not the causal position of the security transition.

Mock middleware events serve a separate purpose. They validate parsing, correlation, missing-field handling, and metric computation. They do not support claims about production routing, policy enforcement, queueing, or backend processing. This evidence separation prevents a test of the analysis chain from becoming surrogate evidence about the regional infrastructure.

2.2.2 Client-Boundary Latency and Saturation Evidence

Iorio, Risso, and Casetti justify the choice of the client boundary by distinguishing network round-trip time from application-level flow completion time. Their end-to-end interval begins with request transmission and ends with complete response reception; it therefore captures transmission, transport, middle-box, application-protocol, and service contributions that a network probe does not expose³. This boundary also avoids the clock-synchronisation requirement of one-way measurements. For a protected regional path, the same principle makes the client the only common observation point across direct and mediated scenarios.

The full-stack interpretation makes Flow Completion Time (FCT) the principal end-to-end timing metric. Iorio et al. show that network round-trip time (RTT) captures only one layer of a latency budget that also contains transmission, forwarding, queueing, transport establishment, application protocol, security, service exposure, orchestration, and computation. RDG extends this observation boundary from one request-response exchange to the terminal outcome of an information-dependent workflow. Request latency remains necessary for locating which transaction population changes, but workflow FCT determines when the domain operation actually completes. This extension transfers the measurement principle, not the numerical results of the edge-computing campaign⁴.

Figure 2.2 makes the measurement hierarchy explicit. The horizontal displacement between the network, transport, and application distributions represents progressively accumulated work rather than three interchangeable latency measures. The curves provide a conceptual reading of the hierarchy analysed by Iorio et al.; they do not reproduce raw samples or RDG observations.

³Marco Iorio, Fulvio Risso, and Claudio Casetti. “When Latency Matters: Measurements and Lessons Learned”. In: *ACM SIGCOMM Computer Communication Review* 51.4 (Oct. 2021), pp. 2–13. DOI: [10.1145/3503954.3503956](https://doi.org/10.1145/3503954.3503956).

⁴[Ibid.](#)

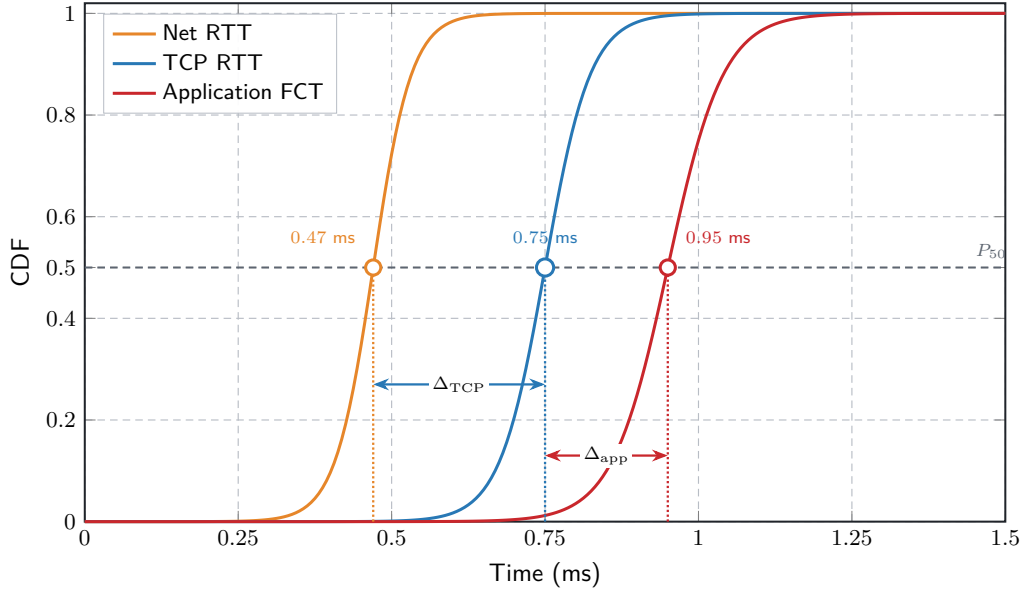


Figure 2.2: Conceptual separation of Network Round-Trip Time (RTT), Transmission Control Protocol (TCP) RTT, and application-level Flow Completion Time (FCT). Median markers expose the additional interval that each observation boundary contains.

For completed request i , the external latency is

$$L_i = t_i^{\text{response}} - t_i^{\text{dispatch}}. \quad (2.3)$$

For matched direct and mediated populations, the primary path estimand is

$$\Delta L = \tilde{L}_{\text{mediated}} - \tilde{L}_{\text{direct}}, \quad (2.4)$$

where $\tilde{L}$ denotes a declared statistic over compatible transaction, payload, outcome, connection-policy, load, and time-window classes. The differential estimates the external effect associated with the path change. It does not decompose the contribution of APMS, transport, protocol transformation, or a downstream service.

Consequently, ΔL represents a *Path-level Overhead*: an aggregated external differential that can contain propagation, transport and security establishment, gateway policy enforcement, Simple Object Access Protocol (SOAP)/ Exten-

sible Markup Language (XML) or Representational State Transfer (REST)/JavaScript Object Notation (JSON) handling, semantic validation, transformation, routing, queueing, and downstream service time. The term *aggregated* is essential. A positive differential supports the claim that the mediated path adds externally observable time under matched conditions; it does not identify which protected internal phase causes that difference. This interpretation follows the multi-layer latency budget in the edge-computing literature while respecting the narrower authorised observation boundary of the regional case. The client boundary becomes informative about saturation only when latency joins other signals. Akram, Akram, and Rehman model distributed load sharing as a race between communication delay and a stochastic availability window. Their analysis links increasing traffic intensity and communication latency to stale state, failed transfers, wasted work, and reduced effective throughput⁵. The load-sharing model does not become a model of the Veneto architecture, and its numerical thresholds do not transfer to Regione Veneto (RVE) transactions. Its methodological contribution is the joint interpretation of latency, state validity, and completed useful work: delay becomes a saturation signal when it coexists with shrinking completion capacity and increasing ineffective activity.

Figures 2.3 and 2.4 translate this contribution into two complementary views. The state model distinguishes useful transitions from a transfer that arrives after the target state loses validity. The distribution view shows the same risk as a narrowing Load-Sharing Window (LSW) when traffic intensity increases. Both views remain methodological abstractions and import neither the paper’s numerical thresholds nor a topology for the regional system.

⁵Saira Akram, Muhammad Asim Akram, and Fattah UR Rehman. “Numerical Evaluation of Network Latency and Throughput in Distributed Systems”. In: *The Science Post* 1.4 (Dec. 2025). DOI: [10.5281/zenodo.18012513](https://doi.org/10.5281/zenodo.18012513).

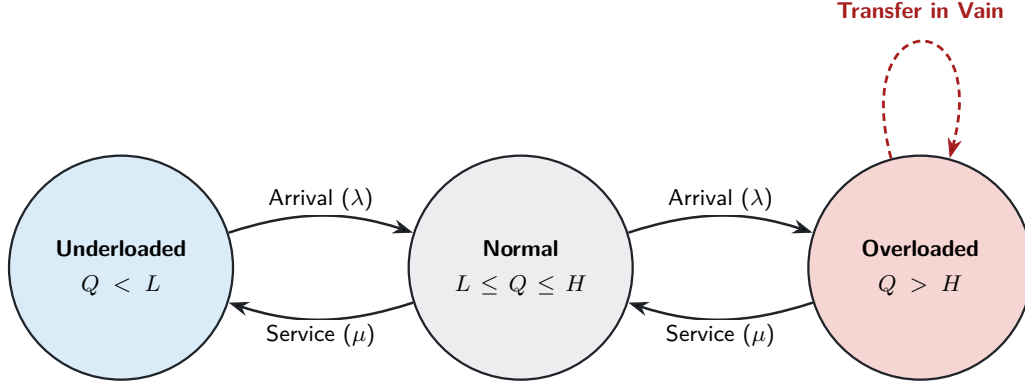


Figure 2.3: Conceptual transitions between underloaded, normal, and overloaded node states. A delayed transfer can become ineffective when it reaches a node whose load state no longer validates the original decision.

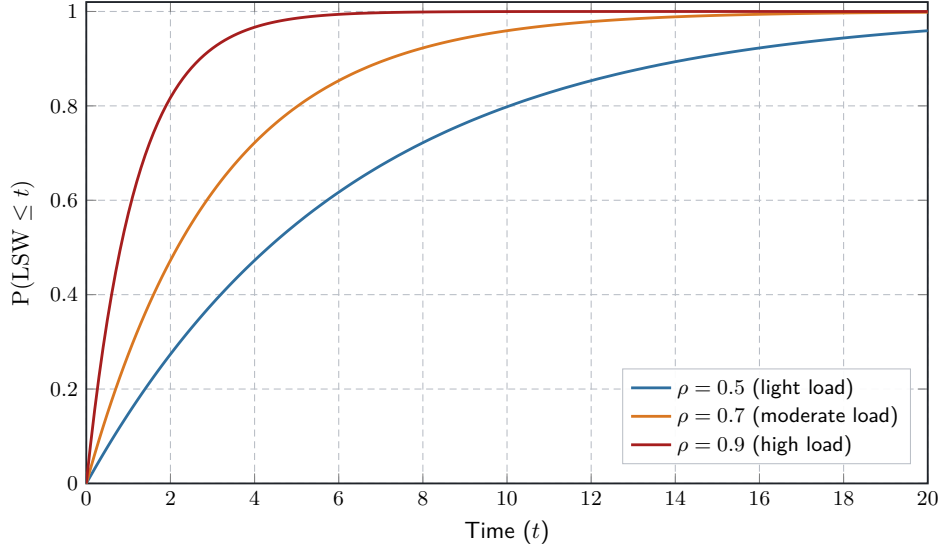


Figure 2.4: Conceptual cumulative distribution of the Load-Sharing Window under increasing traffic intensity. The curves express the qualitative ordering of the model and do not constitute digitised experimental samples.

This stochastic interpretation determines the summary statistics. The median (P_{50}) supplies a robust centre for a skewed latency population, while P_{95} and P_{99} expose the tail in which a delayed prerequisite is most likely to outlive the state that makes it useful. The arithmetic mean remains a supplementary descriptor, but it cannot reveal the probability of crossing a high-latency boundary and it changes markedly under rare delays. Akram et al. justify this

distribution-aware choice through a quantile rule rather than through average latency alone; Iorio et al. likewise evaluate application FCT at high percentiles. RDG therefore reports P_{50} , P_{95} , and P_{99} for a declared homogeneous population and observation window, together with sample size and outcome coverage. The Load-Sharing Window does not represent an actual SI.Ter component: it supplies a conceptual model for the stochastic interval in which information about distributed state remains actionable⁶.

Dimitrov and Skjellum reinforce the need for application-level interpretation. Their experiments show that lower point-to-point latency does not necessarily produce lower application execution time, because communication pattern, message size, completion mode, Central Processing Unit (CPU) overhead, and the possibility of overlapping work determine the complete effect⁷. The studied Message Passing Interface (MPI) environment differs from healthcare interoperability, so the thesis transfers only the evaluation principle: a local latency number cannot stand for the performance of a heterogeneous workflow. Healthcare applications make this principle especially relevant because patient queries, context transitions, document retrieval, and cryptographic signing expose different message sizes, synchronisation points, and blocking dependencies.

An isolated ping-pong benchmark therefore answers a narrower question than the one posed by regional mediation. Dimitrov and Skjellum show that a lower point-to-point latency can coexist with higher processor overhead or reduced opportunity to overlap communication and computation; the regular request-reply pattern also omits the collective and asynchronous structures of real applications. In the healthcare case, an isolated endpoint call omits semantic validation, identity acquisition, branch selection, and failure propagation. RDG is

⁶Akram, Akram, and Rehman, “Numerical Evaluation of Network Latency and Throughput in Distributed Systems”; Iorio, Risso, and Casetti, “When Latency Matters: Measurements and Lessons Learned”.

⁷Rossen Dimitrov and Anthony Skjellum. “Impact of Latency on Applications’ Performance”. Manuscript consulted in the documentary corpus. Apr. 2001. URL: https://users.cs.northwestern.edu/~srg/Papers/01-10-02/srg_2.pdf (visited on 07/31/2026).

consequently necessary as an orchestrator of dependent workflows, because the critical path exists only when each valid output enables the next step. The abstract SAML–JWT–FHIR progression illustrates dependency transfer, but the documented regional model keeps its concrete paths distinct: SAML enables the JWT consumed by RVE-130, whereas a separate SAML-authenticated RVE-54 interaction returns a FHIR result. This separation evaluates real gateway mediation without inventing one unsupported transaction chain⁸.

Figure 2.5 provides an explanatory adaptation of the point-to-point and collective categories in their communication-pattern analysis. The original figures report message counts and sizes rather than sequence traces; the visualisation therefore illustrates the structural distinction without claiming to reconstruct an observed execution.

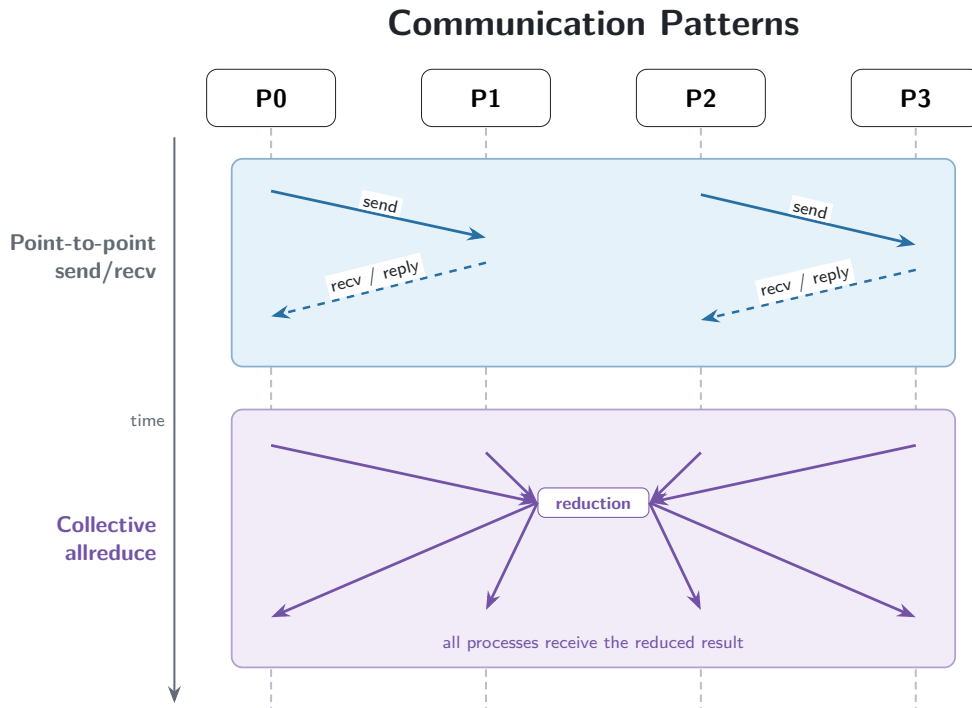


Figure 2.5: Conceptual point-to-point send/receive and collective all-reduce communication patterns, adapted as a sequence-oriented explanation of the categories analysed by Dimitrov and Skjellum.

⁸Dimitrov and Skjellum, “Impact of Latency on Applications’ Performance”.

Latency variability matters for the same reason. A median describes the centre of one population, but a dependent flow reacts to the slowest prerequisite on its critical path. Iorio et al. document dispersion, high-percentile instability, and latency spikes across access, transport, and application conditions; Dimitrov and Skjellum show that the application effect depends on the surrounding communication pattern. In a healthcare workflow, variability can delay the availability of a patient identity, context token, document reference, or signed artefact even when the median request remains stable. The method therefore reports distributions and tails for homogeneous populations and never substitutes a point-to-point mean for complete-flow behaviour.

Saturation is consequently an inference from converging evidence rather than a synonym for high latency. The method requires persistent scheduling drift, occupancy near the configured concurrency bound, a plateau or decline in useful throughput, and a corresponding change in latency, timeouts, or semantic failures. Stable response latency can coexist with increasing delivery drift when the generator no longer dispatches work on time. Conversely, higher latency without a throughput plateau can reflect payload composition, connection establishment, or remote environmental variation rather than saturation.

Table 2.2 fixes the operational meaning of the principal signals.

Table 2.2: Operational measures and their role in saturation analysis.

Measure	Population and unit	Interpretive role
Request latency	Homogeneous completed attempts; milliseconds.	Client-observed dispatch-to-response distribution for one transaction class, reported through P_{50} , P_{95} , and P_{99} with the sample size.

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Measure	Population and unit	Interpretive role
Flow completion	Complete and interrupted occurrences remain separate; milliseconds.	Critical-path effect of orchestration, waiting, and blocking failures.
Useful throughput	Successful terminal flows per declared window.	Completed domain work; it remains distinct from raw request activity.
Request throughput	Completed attempts per declared window.	Technical activity, including calls that belong to unsuccessful flows.
Scheduling drift	Actual minus planned start; milliseconds.	Delivery quality of the generator and early evidence of backpressure.
Outcome distribution	Transport, application, timeout, and terminal-flow classes.	Reliability and preservation of unsuccessful work in every denominator.

The literature therefore supports the measurement boundary and the multi-indicator reasoning, not an imported acceptance threshold. Client-boundary latency supplies a reproducible external signal; ΔL controls the matched path comparison; throughput, drift, concurrency, and outcomes determine whether the signal is consistent with saturation. Internal attribution remains outside the authorised boundary unless a new evidence source enters the protocol.

2.3 Healthcare Workflows as Information Dependencies

2.3.1 Protected Discharge and Anagrafe Zero

Protected discharge gives clinical-process meaning to the profiling strategy. The inpatient unit identifies the patient, prepares assessment information, proposes a destination setting, and forwards the request to the Centrale Operativa Territoriale (COT). The COT evaluates the request and assigns the definitive setting⁹¹⁰. Each stage consumes information that a preceding stage makes available. The workflow is therefore a chain of semantic prerequisites, not a bag of independent service calls.

Patient identity illustrates the first dependency. The project documentation associates protected discharge with patient identification through Anagrafe Zero, while the Anagrafe Zero specification requires RVE-54 Patient Query before other registry transactions¹¹. A successful network response alone does not satisfy this prerequisite. The query has to produce an application outcome and a patient context that the next branch can use. Depending on the profiled outcome, a later operation can assign a new regional identity through RVE-55 or update an existing identity through RVE-57. The branch follows the semantic result of the query; it does not follow only the Hypertext Transfer Protocol (HTTP) status.

This dependency explains why a session-aware orchestrator is necessary. The patient context belongs to one synthetic occurrence and cannot leak into another concurrent occurrence. A failed query blocks the dependent branch, and a warning or application error remains distinguishable from transport success. A generic request generator can reproduce this behaviour only after the test author adds domain-specific state, extraction, validation, and branch rules. The

⁹Raggruppamento Temporaneo di Imprese guidato da AlmavivaA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, pp. 18–21.

¹⁰Azienda Zero, *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*, pp. 35–36.

¹¹Vio et al., *Specifiche tecniche – Anagrafe Zero*.

profiling problem therefore resides in semantic continuity, not in the ability to send a Health Level Seven (HL7) Fast Healthcare Interoperability Resources (FHIR) Release 4 (R4) request.

Semantic FHIR R4 generation is therefore a condition of construct validity, not a convenience for producing payload volume. Anagrafe Zero interprets identifiers, demographic search parameters, resource structure, and transaction outcomes according to its declared FHIR contract. A syntactically valid but semantically arbitrary JSON or XML document can exercise transport while triggering a different validation branch from the represented patient operation. Metadata-driven generation preserves the profile-relevant structure and makes the resulting query, assignment, or update comparable across repetitions. It also keeps the data synthetic, so semantic fidelity does not require personal records¹².

The same principle applies to regional security and context access. An RVE-1.b interaction produces a Security Assertion Markup Language (SAML) assertion; RVE-121 uses the assertion and contextual information to produce the JSON Web Token (JWT) required by RVE-130¹³. The later call remains invalid when the required transition is absent, even if an earlier endpoint returns a nominally successful transport status. Occurrence-local state preserves both the security contract and the cost of acquiring that state.

Figure 2.6 makes the occurrence-local dependencies visible. It keeps the context path and the patient-query path separate: RVE-130 consumes the material produced through RVE-121, whereas RVE-54 consumes the assertion in a distinct FHIR query. This separation prevents the visual model from suggesting one unsupported SAML–JWT–FHIR transaction chain.

Document access adds a further dependency family. Integrating the Healthcare Enterprise (IHE) Cross-Enterprise Document Sharing (XDS.b) separates

¹²Vio et al., *Specifiche tecniche – Anagrafe Zero*.

¹³Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*; Canal and Pavan, *Specifiche tecniche chiamata di contesto RVE*.

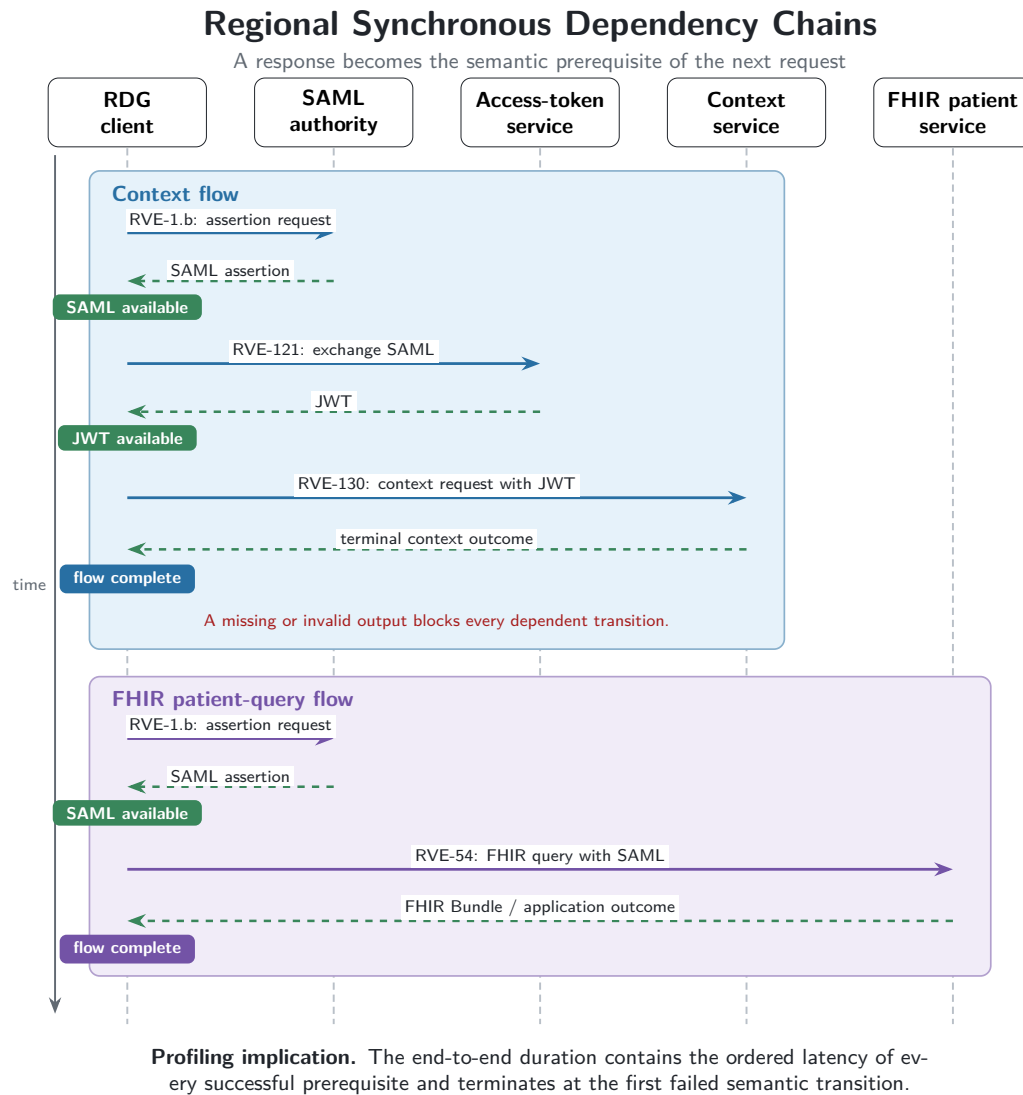


Figure 2.6: Conceptual regional dependency chains retained by the workload model. Each response supplies a semantic prerequisite for the next request within the same occurrence, and a missing prerequisite blocks the dependent transition.

metadata query, document selection, retrieval, and publication¹⁴. An Information Technology Infrastructure (ITI)-18 query returns metadata; an ITI-43 retrieval consumes a repository and document reference; an ITI-41 publication carries document content and metadata. These transactions remain separate performance populations because they differ in payload, outcome, and position in the information chain.

Table 2.3 summarises the dependencies that shape the workload model without claiming that every listed transaction forms one approved protected-discharge implementation.

Table 2.3: Information dependencies retained by the profiling model.

Dependency	Semantic prerequisite	Profiling consequence
Patient identity	RVE-54 produces a usable registry outcome before assignment or update.	The flow branches on application semantics and keeps later steps blocked after failure.
Security identity	RVE-1.b produces occurrence-local assertion evidence.	Protected calls include acquisition cost without retaining the assertion value.
Context session	RVE-121 produces the material required by RVE-130.	Transport success without the required output remains a failed state transition.

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¹⁴IHE International, *IHE IT Infrastructure Technical Framework, Volume 1: Cross-Enterprise Document Sharing (XDS.b), Revision 20.2*; CNR (ICAR) e Dipartimento per la Trasformazione Digitale, *Specifiche tecniche per l'interoperabilità tra i sistemi regionali di FSE: Affinity Domain Italia*.

Dependency	Semantic prerequisite	Profiling consequence
Document access	ITI-18 produces the reference consumed by ITI-43.	Query and retrieval retain separate latency, payload, and outcome populations.
Care transition	Patient identity and assessment precede COT evaluation and destination assignment.	Technical completion does not imply clinical approval or completion of the care pathway.

Both care transitions act as process frames. They identify the initiating actor, decision point, information prerequisites, and terminal organisational outcome. They do not justify an invented chain among every available RVE and ITI transaction. A transaction enters an executable case only when both a technical specification and a process source support its role. This source gate protects the profiling model from turning architectural proximity into an unsupported workflow dependency.

Anagrafe Zero also demonstrates why throughput requires a domain denominator. Three successful endpoint responses do not necessarily represent three completed patient-identity processes. A query can end the flow, enable an assignment, or enable an update; warnings and duplicate-management rules can change the terminal meaning. Request throughput therefore measures technical activity, while flow throughput measures useful completion under an explicit semantic outcome rule.

2.3.2 Cryptographic and Protocol Heterogeneity

Scryba Sign supplies a controlled example of heterogeneous interoperability. The service surface uses SOAP, XML, and Web Services Description Language (WSDL) contracts over protected transport. The synchronous path combines

identity and signer discovery through *GetUserInfo4* with the *SignOneDoc* operation. The document can travel as a Base64-encoded payload, and the service performs cryptographic work and constructs a signed artefact¹⁵. Scryba Sign therefore represents a heterogeneous interoperability constraint, not an optional software module. Treating it as an ordinary document upload removes the protocol, identity, and cryptographic work that the profile has to observe. The methodological question is why the signing path enters the workload, not how the client constructs a particular envelope. A realistic profile has to retain the payload class, signing mode, session semantics, and terminal outcome because each factor changes the critical path. SOAP/XML handling adds representation and envelope costs; Base64 increases the transferred representation relative to the binary source; protected transport adds connection and security work; cryptographic signing adds service-side computation. The Authorised Observation Boundary makes only their aggregate external effect observable as part of the Path-level Overhead.

Signer identity creates a further dependency. The information-discovery step identifies the available signing powers, certificates, and One-Time Password (OTP) devices; the signing request depends on an admissible certificate and on the applicable OTP authorisation path¹⁶. This relationship requires session-aware identity hand-off: certificate selection and OTP state belong to the same signing occurrence and a failed identity prerequisite blocks the cryptographic operation. The complete FCT consequently includes the ordered identity and signing path rather than only the final SOAP response. The current executable perimeter preserves the order *GetUserInfo4–SignOneDoc* for the synchronous population; automated certificate selection and the complete OTP lifecycle remain contract-level extensions, so the thesis does not present them as executed measurements.

The synchronous and asynchronous scenarios define distinct experimental pop-

¹⁵Medas S.r.l., *Scryba Sign 3.x Developer's Guide*, pp. 17–18, 91–96, 107–111.

¹⁶*Ibid.*, pp. 60–62, 91–96.

ulations. In the synchronous scenario, the request-response chain includes the return of the signed document. In the asynchronous scenario, submission produces session and document identifiers, while later status checks and retrieval define completion. The asynchronous path therefore introduces queue residence, polling or notification policy, and a longer state lifecycle¹⁷. Pooling the two modes in one latency distribution destroys construct validity because their terminal events and critical paths differ.

The profiling model consequently stratifies signing evidence by execution mode, payload class, connection policy, and application outcome. The current executable evidence perimeter includes the synchronous signing population. The asynchronous contract remains a separately defined profile and does not acquire results by analogy. This boundary preserves architectural completeness without presenting an unexecuted scenario as observed evidence.

Table 2.4: Methodological separation of Scryba Sign profiling populations.

Profile	Terminal condition	Relevant overhead
Synchronous signing	The signed artefact returns in the synchronous interaction.	SOAP/XML and Base64 transfer, protected transport, cryptographic processing, and response transfer.
Asynchronous signing	Submission state leads to later availability and retrieval of the signed artefact.	Submission, queue residence, status acquisition, retrieval policy, and state lifecycle.

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¹⁷Medas S.r.l., *Scryba Sign 3.x Developer's Guide*, pp. 17–18.

Profile	Terminal condition	Relevant overhead
Large-document path	The signed artefact completes through the applicable document-transfer procedure.	Document size, hashing, external transfer, signing, and retrieval remain explicit strata.

A direct-versus-mediated comparison uses the same signing mode, payload class, application outcome, and connection policy on both paths. Under these controls, ΔL estimates the composite external effect associated with mediation. It does not isolate cryptographic time, SOAP serialisation, policy enforcement, or network transfer. The signing service therefore strengthens the need for surrogate observability: it creates a relevant, heterogeneous critical path while leaving its protected internal decomposition unavailable.

The scenario also confirms the importance of complete-flow metrics. A fast submission does not imply a fast signed-document outcome, just as a successful patient query does not imply completion of a protected-discharge process. The method keeps request latency, state-transition success, and terminal flow completion distinct so that local progress cannot masquerade as domain-level completion.

2.4 Experimental Design and Evidence Semantics

2.4.1 Controls, Variables, and Reproducibility

The primary estimand is the change that RDG observes when a documented operation follows a mediated path instead of a direct path under matched conditions. The independent variable is the selected path. Transaction class, payload class, semantic outcome rule, flow position, offered-load profile, connection policy, concurrency bound, seed policy, duration, and observation window act as controlled variables. Request latency, flow completion, useful throughput, scheduling drift, and outcome distribution act as dependent variables.

Matching requires more than a shared transaction identifier. Direct and mediated populations use compatible payloads, success definitions, flow positions, and workload conditions. Scenario order also remains controlled because connection reuse, cache state, or remote environmental drift can create an apparent path difference. Paired seeds align generated choices, while deterministic shuffling or interleaving reduces a fixed chronology effect. The method retains run timestamps and dispersion because a live remote environment remains variable even when the synthetic workload is reproducible.

Table 2.5 identifies the main confounding sources and the residual limitations that remain after control.

Table 2.5: Controls for an interpretable direct-versus-mediated differential.

Confounding source	Experimental control	Residual limitation
Transaction semantics	Match operation, flow position, and application outcome.	Contract evolution can change equivalence across versions.
Payload and connection state	Stratify payload class and declare connection policy.	Private caches and remote connection management remain opaque.
Offered load	Reuse temporal profile, mix, seed policy, duration, and concurrency.	Unobserved external demand remains uncontrolled.
Execution chronology	Pair repetitions and control scenario order.	Environmental drift can persist within and across pairs.

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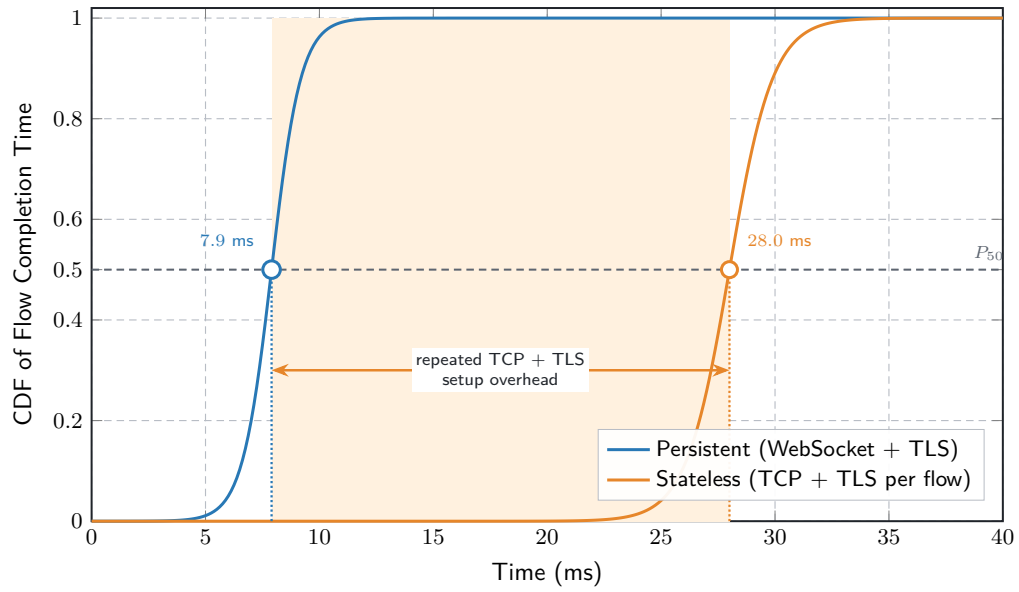
Confounding source	Experimental control	Residual limitation
Generator capacity	Calibrate delivery and retain drift and active-concurrency signals.	Host scheduling remains part of the external experiment.

Every run identifies the workload profile, offered rate, duration, seed, burst parameters, flow mix, concurrency bound, execution mode, configuration state, and artefact version. Reusing a seed aligns paired choices; changing seeds across repetitions samples alternative schedules. Neither action makes the remote environment deterministic. Reproducibility refers to the experimental input and analysis procedure, while environmental repeatability remains an empirical question.

Connection policy deserves explicit control because security setup can dominate short flows. Figure 2.7 gives a conceptual reconstruction of the persistent and stateless populations analysed by Iorio et al. The separation motivates occurrence-local security-state management and declared transport reuse; it does not assert that RDG adopts WebSocket transport or that the reported medians transfer to regional services.

Durations use a monotonic clock. Wall-clock timestamps establish chronology but do not support subtraction across independent hosts without a documented synchronisation bound. Correlation requires compatible identifiers, event semantics, and populations. Missing or unmatched fields remain absent instead of being inferred. These rules prevent timestamp precision from being confused with visibility beyond the authorised boundary.

Structured events retain run, scenario, occurrence, flow, step, and request identity together with planned time, measured duration, and outcome class. Normalisation preserves malformed records, missing terminal events, interrupted flows, and unmatched correlations as data-quality outcomes. Silent removal of



RDG design implication. Preserve occurrence-local security state and reuse transport sessions whenever the transaction contract and the authorised security policy permit it.

Figure 2.7: Conceptual FCT comparison between a persistent secure connection and a stateless flow that repeats transport and Transport Layer Security (TLS) establishment. The differential motivates explicit connection policy and occurrence-local security-state management in RDG.

these records biases both latency and reliability toward successful observations. Provenance therefore precedes performance interpretation.

2.4.2 Evidence Gates and Ex Ante Judgement

Every performance statement requires an operational definition, a population, a unit, an aggregation level, and an event source. If $C(w)$ denotes the number of completed units in a window w , then

$$X(w) = \frac{C(w)}{|w|} \quad (2.5)$$

defines throughput only after the method declares whether $C(w)$ counts requests, terminal flows, or successful terminal flows. Retries can increase request throughput without increasing useful flow throughput, and early failures can reduce request activity while increasing the number of incomplete flows.

Criteria remain fixed before the corresponding results enter the judgement. For criterion j , the thesis uses

$$C_j = (S_j, P_j, m_j, s_j, w_j, d_j, \tau_j, E_j), \quad (2.6)$$

where S_j is the scenario, P_j the population, m_j the metric, s_j the statistic, w_j the window, d_j the preferred direction, τ_j an externally justified threshold, and E_j the evidence gate. A numerical value is not assessable when its population, boundary, or threshold provenance is missing.

Table 2.6: Ex ante criteria and required evidence.

Criterion	Population	Required evidence
Latency	Homogeneous requests and complete or interrupted flows remain separate.	Timer boundary, statistic, sample size, window, repetitions, path, and outcome rule.

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Criterion	Population	Required evidence
Throughput	Request and useful-flow completions under declared offered load.	Profile, mix, concurrency, calibration, duration, and plateau rule.
Reliability	Transport, application, timeout, and terminal-flow outcomes.	Error taxonomy, denominator, retry rule, and semantic outcome rule.
Saturation	Drift, active concurrency, throughput, latency, and failures.	Repeated convergence of indicators rather than offered rate alone.
Data quality	Expected, parsed, malformed, incomplete, and unmatched events.	Event semantics, provenance, coverage rule, and completeness threshold.

Observation completeness conditions every statistic. The configured timeline and flow definition determine expected requests and terminal events; the retained event stream determines parsed observations. A missing end event, duplicate identifier, malformed record, or unmatched occurrence changes the admissible population. Successful-request latency therefore appears beside timeouts and application failures rather than replacing them.

The judgement vocabulary follows the evidence hierarchy. A criterion is *not assessable* when its threshold or evidence gate is absent; *conditionally supported* when structural or controlled evidence exists without the required live coverage; *accepted* when valid evidence meets an externally declared rule; and *rejected* when valid evidence violates that rule. Mock evidence supports instrument validation. Live external evidence supports observations for the declared en-

vironment. Internal causal evidence requires a newly authorised source inside the protected boundary.

Threshold provenance remains external to the campaign under judgement. An approved service-level requirement, contractual criterion, or preregistered protocol can supply τ_j ; an observed percentile cannot become its own pass value. The Sistema Informativo Territoriale (SI.Ter) procurement specification supplies a bounded front-end reference for interactions that do not depend on third-party systems¹⁸. That reference does not automatically transfer to mediated RVE workflows because their dependencies and observation boundary differ.

Repeated scenarios reduce but do not remove temporal confounding. Paired seeds align the synthetic choices, controlled ordering limits a fixed sequence effect, and run timestamps retain the chronology needed to detect environmental change. Remote observations within one run remain dependent. The final judgement therefore considers run-level dispersion and consistency of direction instead of relying only on a pooled request distribution.

Negative evidence remains part of the decision. A malformed event, missing terminal state, failed token transition, timeout, or generator overload can make a performance criterion not assessable while still exposing a framework or contract problem. Data-quality and semantic gates apply before aggregation, and their failures remain visible in the reported denominator.

2.5 Generic versus Domain-Specific Profiling

2.5.1 Modelling Distance and Semantic Fidelity

The comparison between JMeter and RDG concerns paradigms, not product feature counts. Apache JMeter represents a mature generic request-centric paradigm: its test plans combine samplers, controllers, extractors, variables,

¹⁸Azienda Zero, *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*, pp. 56–57.

assertions, scripts, and distributed execution mechanisms¹⁹. RDG represents a domain-specific workflow paradigm: its primary abstraction is a typed health-care flow whose security, data, dependency, scheduling, and evidence semantics remain part of one executable model.

Both paradigms can issue concurrent SOAP/XML and REST/JSON requests. Raw protocol reach is therefore not the deciding criterion. The deciding criterion is *modelling distance*: the amount of additional interpretation and custom state logic required to move from the experimental model to the documented regional workflow. In a generic paradigm, the author composes scoped variables, extractors, assertions, controllers, and scripts to preserve SAML propagation, patient-query outcomes, context transitions, document references, and failure blocking. In a domain-specific paradigm, these relationships constitute the workflow vocabulary itself.

This distinction does not make the generic paradigm technically incapable. It makes semantic fidelity an author-maintained convention distributed across plan elements. The risk is not only maintenance cost. A test can remain executable while becoming semantically invalid: state can leak across virtual users, a dependent request can start after a nominal transport success, or a failed prerequisite can disappear from the useful-flow denominator. Reducing modelling distance reduces the number of places in which these inconsistencies can arise.

The domain-specific paradigm also connects workload construction and evidence construction. Domain-specific metadata generation creates synthetic patient and document contexts; stateful orchestration logic preserves occurrence-

¹⁹Apache Software Foundation. *Apache JMeter User's Manual: Elements of a Test Plan*. URL: https://jmeter.apache.org/usermanual/test_plan.html (visited on 08/04/2026); Apache Software Foundation. *Apache JMeter User's Manual: Component Reference*. URL: https://jmeter.apache.org/usermanual/component_reference.html (visited on 08/04/2026); Apache Software Foundation. *Apache JMeter User's Manual: Functions and Variables*. URL: <https://jmeter.apache.org/usermanual/functions.html> (visited on 08/04/2026); Apache Software Foundation. *Apache JMeter User's Manual: Remote (Distributed) Testing*. URL: <https://jmeter.apache.org/usermanual/remote-test.html> (visited on 08/04/2026).

local outputs and branches; the asynchronous execution model expresses concurrency among independent flows; the event model retains both technical attempts and terminal domain outcomes. The resulting artefact translates documented healthcare dependencies into executable and observable experimental units.

The comparison in Table 2.7 avoids implementation names.

Table 2.7: Generic and domain-specific profiling paradigms. JMeter represents the generic request-centric paradigm, while RDG represents the domain-specific workflow paradigm.

Dimension	Generic request-centric paradigm	Domain-specific workflow paradigm
Unit of analysis	Request or sampler sequence extended with user-defined state.	Typed information-dependent healthcare flow.
State semantics	Scope and lifetime emerge from variables, extractors, controllers, and custom logic.	Occurrence-local state and prerequisite blocking form explicit invariants.
Data semantics	Payloads remain generic unless the plan incorporates domain-specific generation and validation.	Synthetic patient and document metadata preserve the represented profile without using personal data.

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Dimension	Generic request-centric paradigm	Domain-specific workflow paradigm
Outcome semantics	Transport assertions require additional domain rules for application and flow completion.	Transport, application, and terminal-flow outcomes remain separate by construction.
Observability	Endpoint timing is native; workflow correlation and path differentials require explicit composition.	Correlated external events directly support surrogate path observation.
Evolution cost	Broad ecosystem and mature distributed facilities; domain contracts remain test-plan responsibilities.	Lower modelling distance for represented contracts; domain model requires maintenance when specifications change.

The comparison therefore concerns semantic risk and experimental traceability. A generic plan is attractive when the target is a stateless endpoint or a protocol-level baseline. A domain-specific workflow is attractive when the experimental outcome depends on ordered state transitions, domain metadata, application-level validation, and flow-level denominators. The Veneto use case belongs to the second class because security, identity, context, document, and signing transitions determine whether useful work completes.

2.5.2 Ex Ante Justification of RDG

RDG is methodologically justified when five conditions hold: the target process contains dependencies whose violation changes experimental meaning; authorised evidence excludes production traces and internal regional telemetry; synthetic data preserve contract structure without protected material; the research question concerns complete-flow behaviour and path differentials; and one evidence model connects offered load, state transitions, external timing, and terminal outcomes. The regional case satisfies all five conditions.

RDG does not claim that its execution engine is faster. A throughput comparison with JMeter requires matched workloads, equal host conditions, and a dedicated generator benchmark, which lies outside this chapter. A stateless generic baseline remains useful for calibration and protocol isolation, but it does not replace a session-aware workload for protected regional workflows.

The choice relocates rather than eliminates complexity. A domain-specific model requires versioned transaction semantics, outcome definitions, scheduling controls, and event provenance. The same elements otherwise reappear as dispersed custom logic in a generic plan; RDG makes them inspectable at the abstraction level of the research question.

Scientific Rationale for Domain-Specific Profiling.

RDG responds to latency-aware distributed systems by combining quantile-based stochastic interpretation, full-stack FCT, matched Path-level Overhead, application-representative communication patterns, metadata-driven FHIR R4 generation, and session-aware orchestration. The absence of authorised traces makes controlled synthetic workflows the admissible internally valid input, while the regional security perimeter imposes Advanced Black-box with Surrogate Observability. RDG is therefore not merely custom software: it translates international methodological consensus into Regione Veneto's operational, semantic, and security constraints. Its evidence supports reproducible external profiling and workflow inference, not internal causes or unexecuted branches.

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